Notes on Template Games and Differential Logic

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1 General Remarks

The results from [FGHW08] introduce the bicategorical interpretation of linear logic based on distributors and generalised species. The idea is that every formula A of linear logic is interpreted as a small category $[\![A]\!]$ and every derivation tree

$$\dfrac{\begin{array}{c}\pi \\ \dots\end{array}}{A_1, \dots, A_n \vdash A}$$

is interpreted as a distributor (or a profunctor)

$$[\![\pi]\!] : [\![A_1]\!] \otimes \dots \otimes [\![A_n]\!] \not\rightarrow [\![A]\!].$$

Recall that a distributor $M : A \not\rightarrow B$ is defined as a functor $M : A \times B^{op} \to \mathbf{Set}$. The category $\mathbf{Dist}$ of small categories and distributors is symmetric monoidal with the tensor product defined by the direct product of categories in $\mathbf{Cat}$, the category of small categories and functors, and the unit defined by the terminal (a single-element) category. $\mathbf{Dist}$ is also *-autonomous (and, moreover, compact closed) with linear negation defined as $A \mapsto A^{op}$. We thus have the following isomorphism

$$\mathbf{Dist}(A \otimes B, C) \cong \mathbf{Dist}(B, A^{op} \otimes C)$$

natural A, B and C, providing the currification in $\mathbf{Dist}$.

The exponential modality $A \mapsto\, !\,A$ is interpreted using the 2-monad $\mathbf{Sym} : \mathbf{Cat} \to \mathbf{Cat}$ sending each small category $\mathcal{C}$ to the free symmetric monoidal category generated by $\mathcal{C}$. $\mathbf{Sym}(\mathcal{C})$ has objects defined as finite sequences $A_1 \dots A_n$ for objects from $\mathcal{C}$ for where $A_i \in \mathcal{C}$ for $1 \leq i \leq n$, and morphisms

$$(\sigma, f_1 \dots f_n) : A_1 \dots A_n \to B_1 \dots B_n$$

defined as pairs $(\sigma, f_1 \dots f_n)$ consists of a permutation $\sigma \in \mathcal{S}_n$ on the set of n-elements along with a sequence of morphisms from $\mathcal{C}$ for $1 \leq k \leq n$

$$f_k : A_k \to B_{\sigma(k)}.$$

The category $\mathbf{Sym}(\mathcal{C})$, as strict monoidal, comes with a pair of functors $\otimes : \mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C}) \to \mathbf{Sym}(\mathcal{C})$ and $I_A : \mathbb{1} \to \mathbf{Sym}(\mathcal{C})$ defined as the concatenation and the empty sequence respectively. $\mathbf{Sym}(\mathcal{C})$ with those two functors also forms a monoid in $\mathbf{Cat}$, and such a monoid is commutative up to isomorphism γ (the symmetry of the monoidal category):

$$\begin{tikzcd}
\mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C}) \arrow[rr, "(21)"] \arrow[dr, "\otimes"'] & \arrow[d, phantom, "\Rightarrow_{\alpha}"] & \mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C}) \arrow[dl, "\otimes"] \\
& \mathbf{Sym}(\mathcal{C}) &
\end{tikzcd}$$

where (21) is the symmetry of the cartesian category $\mathbf{Sym}(\mathcal{C}) \times \mathbf{Sym}(\mathcal{C})$. The exponential modality $A \mapsto {!}\,A$ is, thus, interpreted in $\mathbf{Dist}$ by turning $\mathbf{Sym}$ into a 2-comonad

$$\mathbf{Sym} : \mathbf{Dist} \to \mathbf{Dist}$$

using the distributivity law between $\mathbf{Sym}$ and the presheaf construction in $\mathbf{Cat}$. Further, every functor $F : \mathcal{C} \to \mathcal{D}$ induces an adjoint pair $L_F \dashv R_F$ of distributors $L_F : A \not\rightarrow B$ and $R_F : B \not\rightarrow A$ in the bicategory $\mathbf{Dist}$, where the distributors L_F and R_F are given as the following presheaves

$$\begin{array}{c} L_F(B, A) = \mathcal{D}(FB, A) : B^{op} \times A \to \mathbf{Set} \\ R_F(A, B) = \mathcal{D}(A, FB) : A^{op} \times B \to \mathbf{Set} \end{array}$$

The comonoid structure $(\mathbf{Sym}(\mathcal{C}), d_{\mathcal{C}}, e_{\mathcal{C}})$ of the exponential modality $\mathbf{Sym}$ is given by the following distributors

$$\begin{array}{c} d_{\mathcal{C}} = R_{\otimes_{\mathcal{C}}} : \mathbf{Sym}(\mathcal{C}) \not\rightarrow \mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C}) \\ e_{\mathcal{C}} = R_{I_{\mathcal{C}}} : \mathbf{Sym}(\mathcal{C}) \not\rightarrow \mathbb{1} \end{array}$$

right adjoint to $\otimes_{\mathcal{C}}$ and $I_{\mathcal{C}}$ respectively. Note that $(\mathbf{Sym}(\mathcal{C}), d_{\mathcal{C}}, e_{\mathcal{C}})$ is not commutative, but only symmetric, so it is commutative only up to isomorphism:

$$\begin{tikzcd}
& \mathbf{Sym}(\mathcal{C}) \arrow[dl, "d_{\mathcal{C}}"'] \arrow[dr, "d_{\mathcal{C}}"] \arrow[d, phantom, "\Rightarrow_{\alpha}"] & \\
\mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C}) \arrow[rr, "(21)"] & & \mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C})
\end{tikzcd}$$

where γ is the symmetry of $\mathbf{Sym}(\mathcal{C}) \otimes \mathbf{Sym}(\mathcal{C})$, which is a distributor in $\mathbf{Dist}$ as well.

The necessity to weaken the commutativity condition to an isomorphism is similar to the phenomenon of the Seely isomorphism in linear logic:

$$!(A \& B) \cong {!}\,A \otimes {!}\,B$$

One can interpret this equation of linear logic as a distributor with the concatenation functor:

$$\mathrm{concat}_{A,B} : \mathbf{Sym}(A) \times \mathbf{Sym}(B) \to \mathbf{Sym}(A + B)$$

that takes sequences $u = a_1 \dots a_p$ and $v = b_1 \dots b_q$ of objects from A and B respectively concatenates them into the sequence of objects from $A + B$:

$$\mathrm{concat}_{A,B}(u, v) = a_1 \dots a_p \cdot b_1 \dots b_q$$

The distributor interpreting the Seely isomorphism

$$\mathbf{Sym}(A \& B) \nrightarrow \mathbf{Sym}(A \otimes B)$$

is defined as the right adjoint of the concatenation functor where $A\&B$ stands for $A + B$ for the disjoint union of A and B. Besides, $\mathrm{concat}_{A,B}$ is equivalence of categories injective on objects, but it is not an isomorphism. Thus we replace here the Seely isomorphism with the Seely equivalence.

2 On Models of Differential Linear Logic

A good introduction to differential linear logic is [Ehr18]. If we interpret the exponential modality as $A \mapsto \mathbf{Sym}(A)$, then one can see that the model of distributors is also a model of differential linear logic since the exponential modality $\mathbf{Sym}(A)$ comes with the monoid structure in **Dist**:

- $m_A = L_{\otimes_A} : \mathbf{Sym}(A) \otimes \mathbf{Sym}(A) \nrightarrow \mathbf{Sym}(A)$,
- $u_A = L_{I_A} : \mathbb{1} \nrightarrow \mathbf{Sym}(A)$.

whose multiplication and unit are the left adjoint distributors associated to the monoid structure of $\mathbf{Sym}(A)$ in **Cat**. Note that the monoid $(\mathbf{Sym}(A), m_A, u_A)$ and the comonoid $(\mathbf{Sym}(A), d_A, e_A)$ are the left and right adjoint "avatars" of the monoid structure $(\mathbf{Sym}(A), \otimes_A, I_A)$ in **Cat**. The monoid and the comonoid structures on $\mathbf{Sym}(A)$ define a bialgebra, but the bialgebra structure is only up to an invertible natural transformation:

$$\begin{tikzcd}
& & \mathbf{Sym}(A) \arrow[drr, "d_A"] \arrow[dd, phantom, "\Downarrow \sim"] & & \\
\mathbf{Sym}(A)^{\otimes 2} \arrow[urr, "m_A"] \arrow[dr, "d_A \otimes d_A"'] & & & & \mathbf{Sym}(A)^{\otimes 2} \\
& \mathbf{Sym}(A)^{\otimes 4} \arrow[rr, "(1324)"'] & & \mathbf{Sym}(A)^{\otimes 4} \arrow[ur, "m_A \otimes m_A"'] &
\end{tikzcd}$$

The explanation is that the disjoint sum $A + B$ of small categories A and B coincide with their cartesian sum $A \oplus B$ and their cartesian product $A\&B$. Therefore, every object A is equipped with a monoid and a comonoid structure:

- $\nabla_A : A \oplus A \nrightarrow A$,
- $\nabla_A^0 : 0 \nrightarrow A$,
- $\Delta_A : A \nrightarrow A \oplus A$,
- $\Delta_A^0 : A \nrightarrow 0$.

defined as the left and right adjoint distributors to the canonical functors $A + A \to A$ and $0 \to A$, where 0 is the initial category in **Cat**. The multiplication ∇_A and the comultiplication Δ_A are equipped with a bialgebra structure in **Dist**, up to an invertible natural transformation:

$$\begin{tikzcd}
 & & A \arrow[drr, "\Delta_A"] \arrow[dd, Rightarrow, "\sim"] & & \\
A^{\oplus 2} \arrow[urr, "\nabla_A"] \arrow[dr, "\Delta_A \oplus \Delta_A"'] & & & & A^{\oplus 2} \\
 & A^{\oplus 4} \arrow[rr, "(1324)"'] & & A^{\oplus 4} \arrow[ur, "\nabla_A \oplus \nabla_A"'] &
\end{tikzcd}$$

The exponential modality $A \mapsto \mathbf{Sym}(A)$ along the family of Seely equivalences

$$\mathbf{Sym}(A^{\oplus n}) \nrightarrow (\mathbf{Sym}(A))^{\otimes n}$$

defines a lax and oplax symmetric monoidal bifunctor

$$\mathbf{Sym} : (\mathbf{Dist}, \oplus, 0) \to (\mathbf{Dist}, \otimes, \mathbb{1})$$

The differential in DiLL is interpreted the following way in the category of distributors. One starts with a family of functors

$$\eta_A : A \to \mathbf{Sym}(A)$$

defining the unit of the monad $A \mapsto \mathbf{Sym}(A)$. One can postcompose the functor with $\otimes_A$ to obtain the functor:

$$\mathrm{append}_A : (a, v) \mapsto a \cdot v \in \mathbf{Sym}(A) : A \times \mathbf{Sym}(A) \to \mathbf{Sym}(A)$$

The differential is interpreted in **Dist** as the left adjoint distributor

$$\partial_A : A \otimes \mathbf{Sym}(A) \nrightarrow \mathbf{Sym}(A)$$

associated to the append_A, whereas the dereliction and the codereliction morphisms:

- $\mathrm{der}_A : \mathbf{Sym}(A) \nrightarrow A$,
- $\mathrm{coder}_A : A \nrightarrow \mathbf{Sym}(A)$

are associated to η_A.

References

[Ehr18] Thomas Ehrhard. An introduction to differential linear logic: proof-nets, models and antiderivatives. *Mathematical Structures in Computer Science*, 28(7):995–1060, 2018.

[FGHW08] Marcelo Fiore, Nicola Gambino, Martin Hyland, and Glynn Winskel. The cartesian closed bicategory of generalised species of structures. *Journal of the London Mathematical Society*, 77(1):203–220, 2008.